

WASP-36b

R-2288

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_130316 · created 2026-08-17T00:41:05+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456367.73019 +/- 0.00053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.026 +/- 0.046
Transit depth (Rp/Rs)^2	0.066 +/- 0.24 [%]
Orbital Inclination (inc)	79.42 +/- 2.96
Airmass coefficient 1 (a1)	1.64 +/- 0.27
Airmass coefficient 2 (a2)	-0.31 +/- 0.18
Scatter in the residuals of the lightcurve fit is	17.58 %
Best Comparison Star	#3 - [456, 125]
Optimal Aperture	1.5
Optimal Annulus	10.47
Transit Duration (day)	0.022 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.026 ± 0.046 vs 0.1368 (deviation 2.03σ)
Duration fit vs published	0.022 d vs 0.0758 d (deviation 1.68σ)
Mid-time O–C	-0.7 min
Depth SNR	0.5
Residual scatter	17.58 %

Light curve

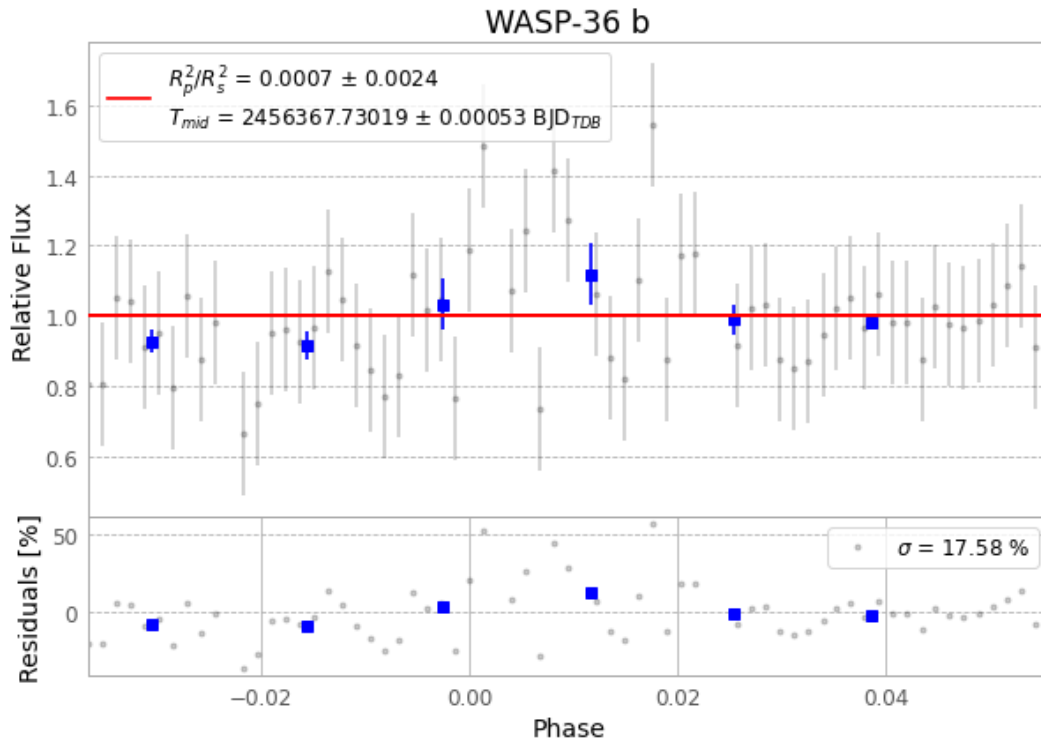


Figure 1 — FinalLightCurve_WASP-36 b_2013-03-15.png (SHA-256 f92723cd6fbd82a6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 316], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 452e6f39d9502616...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

69 FITS frames (45 MB), WASP-36130316040312.FITS → WASP-36130316072712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-130316.log (355ebe695e96...), FinalLightCurve_WASP-36 b_2013-03-15.png (f92723cd6fbd...), FinalLightCurve_WASP-36 b_2013-03-15.csv (e4edb42e513e...)

Compute resources

Wall time 123.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 00:41Z.